

ETHAN ALDRICH

Chief Technology Officer · Software, Infrastructure & Security Engineering

Full-Stack Development · Platform Engineering · Network & IT Operations

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SUMMARY

Technology leader and hands-on engineer responsible for the full IT, network, and software footprint of two Las Vegas companies: a rental car operator with eight locations, and a management group with nine subsidiaries. Comfortable end to end: Cisco IOS and VLAN design, Kubernetes and CI/CD, and the Node.js and C#/.NET applications that replaced a paper-driven business with software I wrote. Currently pursuing a B.S. in Cybersecurity and Information Assurance, with CompTIA A+, Network+, IT Operations Specialist, and ITIL v4 in hand.

EXPERIENCE

Chief Technology Officer

Sep 2021 – Present · Las Vegas, NV

Malco Enterprises of Nevada · Budget Rent A Car of Las Vegas

- Lead technology strategy, infrastructure, and cybersecurity operations across eight rental locations serving the Las Vegas metropolitan area.
- Own the network end to end — SD-WAN, Cisco IOS, VLAN segmentation, PoE, firewall policy, VPN infrastructure, and intrusion monitoring — and maintain PCI DSS compliance across all payment processing systems.
- Architected and operate the distributed application stack that replaced paper processes company-wide: digital check-in/check-out (~5,600 transactions/day), a claims management platform with vehicle lifecycle tracking, digital vehicle inspections, and a rental workflow automation engine — Node.js/Express/MongoDB behind Nginx with load balancing on DigitalOcean Kubernetes.
- Built Maverik (C#/.NET), an API wrapper over the mainframe rental platform, improving operational efficiency by 85%.
- Deployed and maintain six self-hosted GitHub Actions runners across two GitHub organizations for parallel CI/CD across all internal repositories.
- Run the internal Python service tier: FastAPI APIs, Celery/Redis job queues monitored with Flower, and SQLAlchemy/Alembic schema migrations, with cloud infrastructure managed as code in Terraform.
- Built internal Model Context Protocol (MCP) servers putting the rental, fleet-reporting, and rate-management SQL Server databases behind an agent-callable interface.
- Administer Active Directory (provisioning, Group Policy, access control), Windows and Linux servers, SQL Server with Power BI reporting, the Wizard and WAND rental platforms, and a GLPI-based ITSM practice.

Chief Technology Officer

Nov 2021 – Present · Las Vegas, NV

Twelve Management

- Built the complete IT and network infrastructure from the ground up for a management company and nine subsidiaries spanning development, jet charter, and retail.
- Directed cross-functional technical operations for a team of 52, holding system reliability, security, and scalability through a period of rapid organizational growth.
- Established standardized technology practices, security policies, and operational procedures across all subsidiary entities.

Assistant Store Manager

Jun 2017 – Sep 2021 · Las Vegas, NV

Starbucks Coffee Company

- Led shift operations, scheduling, inventory, and staff training for a high-volume retail location.

PROJECTS & OPEN SOURCE

Montr — Distributed Digital Signage · Rust, Node.js/TypeScript · 2025 – Present

Rust playback client and a Node/TypeScript control server driving up to 25 concurrent displays over WebSocket. Cross-platform builds (Linux, macOS, Windows) with an automated multi-platform release pipeline in GitHub Actions. Running in production across Budget lobby displays.

Internal MCP Tooling · Python, Model Context Protocol, SQL Server · 2025 – Present

Model Context Protocol servers exposing the Wizard rental, fleet-reporting, and rate-management SQL Server databases to AI agents through a typed tool interface: object and schema search, query explain plans, per-database health checks, and SQL execution — so operational questions get answered without handing out direct database access.

Yacht Network & AV Integration · 2025

Full network and AV build on an Ocean Alexander 32E and a Princess Y72: UniFi switching and Wi-Fi, VLAN segmentation, PoE, Control4 integration, and dual-Starlink failover.

Proxmox Homelab · 2020 – Present

Multi-node Proxmox VE cluster running dozens of self-hosted services behind Nginx and Tailscale — the proving ground for most of what later ships at work.

ethanaldrich.org · Next.js 16, React 19, TypeScript

Personal site built as an interactive Tokyo-Metro subway map: config-driven data layer, pure-TypeScript SVG geometry with no DOM measurement, a 90% unit-coverage gate, and Playwright end-to-end tests across five browser targets.

SKILLS

Infrastructure & Cloud: Kubernetes · Docker · Proxmox VE · DigitalOcean · S3-compatible object storage · Nginx (reverse proxy, load balancing) · Linux (Ubuntu/Debian/Arch) · Windows Server

Networking: SD-WAN · Cisco IOS · VLAN segmentation · UniFi · Firewalls · VPN · DNS/DHCP · PoE · Tailscale · Starlink failover

Security & Compliance: PCI DSS · network segmentation · firewall and VPN policy · intrusion monitoring · Active Directory access control · TLS/SSL · GitHub Security (CodeQL, Dependabot, secret scanning) · GPG-signed commits · security auditing

Languages: TypeScript · JavaScript · Python · C# · Rust · C/C++ · SQL (T-SQL) · Bash · PowerShell · HTML/CSS

Frameworks & Runtimes: Node.js · Express · React · Next.js · Astro · FastAPI · .NET · TailwindCSS · Vite

Data & Queues: PostgreSQL · MongoDB · Microsoft SQL Server · SQLite · Redis · Celery · Flower · SQLAlchemy · Alembic · Power BI

Architecture: Microservices · event-driven architecture · distributed systems · async job queues · REST API design · CI/CD pipeline design

DevOps & Automation: GitHub Actions (self-hosted runners, multi-org) · Dependabot · branch protection · semantic versioning · multi-platform release automation · Vitest and Playwright test automation

APIs & Integrations: Model Context Protocol (MCP) servers · Microsoft Graph API · DigitalOcean API · RingCentral · NetSuite · Mailcow

IT Operations & Leadership: ITIL v4 · GLPI ITSM · Active Directory and Group Policy · technology procurement and vendor evaluation · cross-functional team leadership

EDUCATION

Western Governors University — B.S. Cybersecurity and Information Assurance

Expected May 2029

M.S. Cybersecurity and Information Assurance planned.

West Career and Technical Academy — High School Diploma, Graphic Design · Las Vegas, NV

2015 – 2020

CERTIFICATIONS

CompTIA A+ · CompTIA Network+ · CompTIA IT Operations Specialist · ITIL v4 Foundations